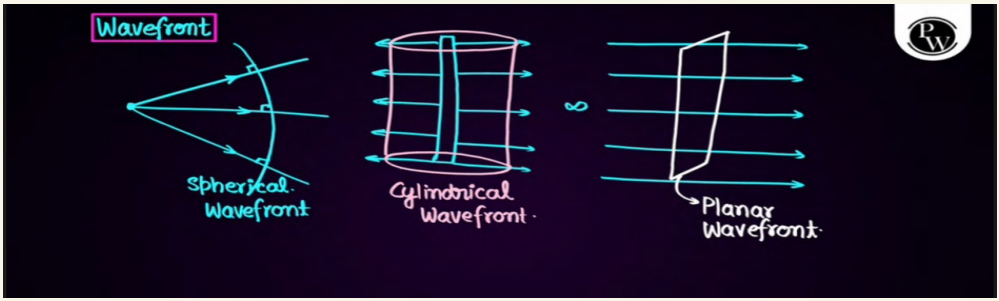
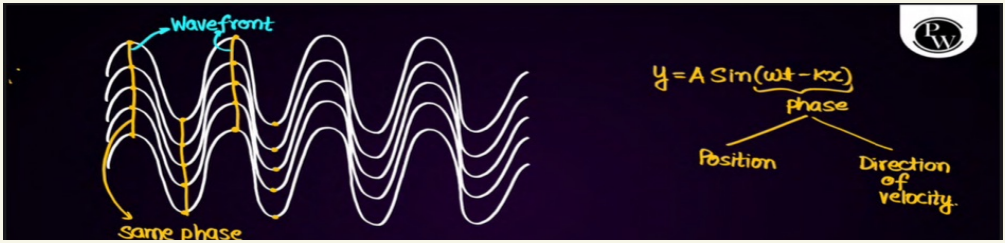
12th hacker series

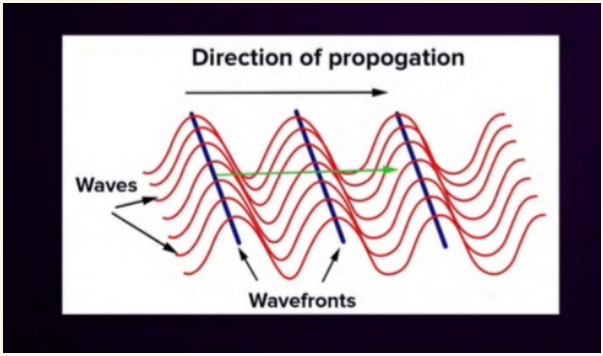
Wave optics

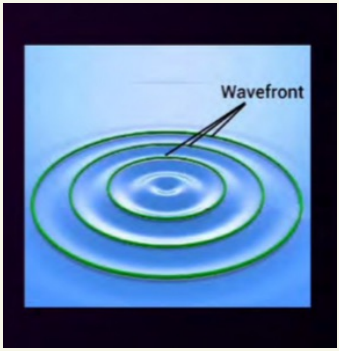
Wavefront





- the locus of all the particles of the medium, which at any instant are vibrating in the same phase is called the wavefront
- thus , a wavefront is a surface of constant phase
- The speed with which the wavefront moves outward from the source is called the phase speed





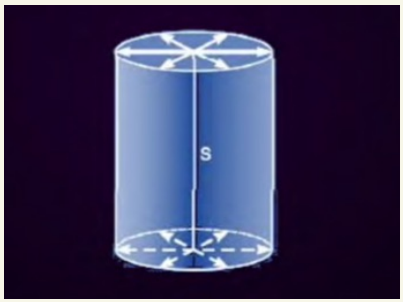
Spherical wavefront

- a spherical wavefront is produced by a point source of light
- it is because the locus of all such points which are equidistant from the point source is a sphere as shown



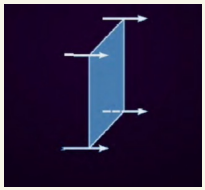
Cylindrical wavefront

- when the source of light is linear in shape, a cylindrical wavefront is produced
- it is because all the points which are equidistant from the linear source lie on the surface of the cylinder



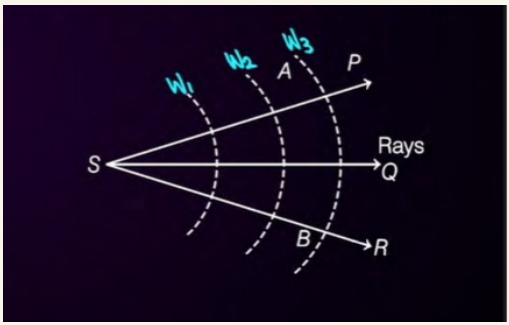
Plane wavefront

- a small part of a spherical or a cylindrical wavefront originating from a distant source will appear plane and hence it is called a plane wave front as shown



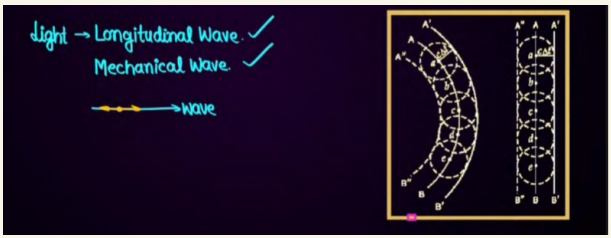
Ray of light

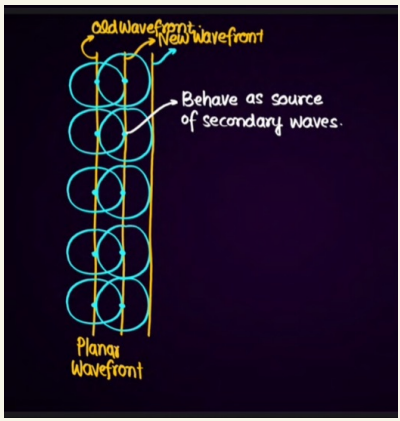
- an arrow drawn normal to the wavefront, and pointing in the direction of propagation of disturbance, represent a ray of light
- a ray of light is the path along which light travels
- Since the ray of light is normal to the wavefront, it is sometimes called as wave normal



Huygen's principle of secondary wavelets

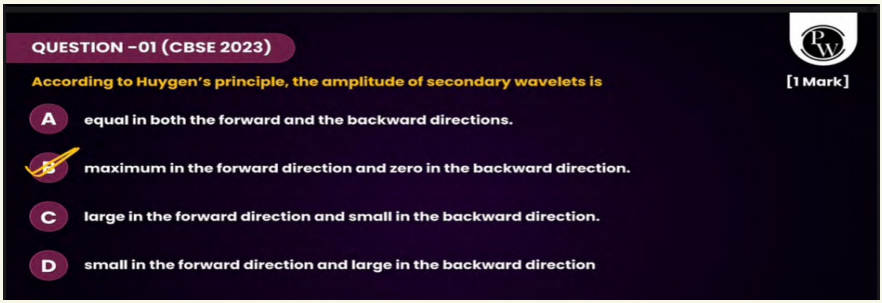
- Huygens principal is the basis of wave theory of light
- it tells how a wavefront propagates through a media
- Each point on a wavefront is a source of secondary waves, which add up to give a wavefront at any later time

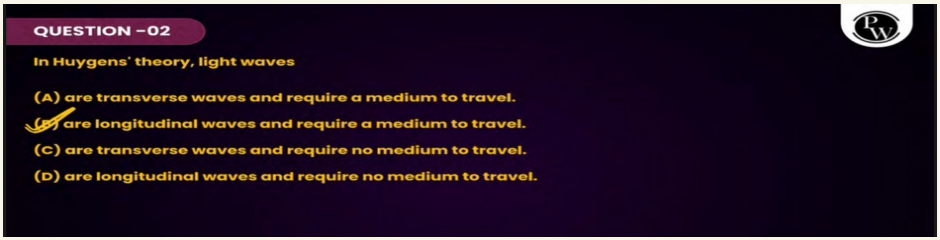




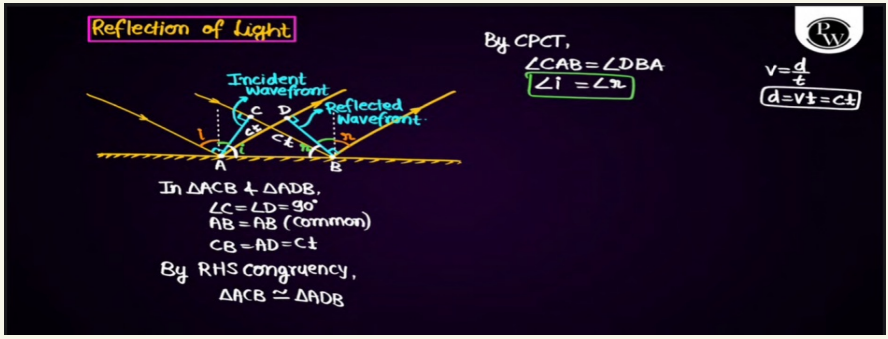
- this principle is based on the following assumptions
- 1) each point on a wavefront act as a fresh source of a new disturbance called secondary waves or wavelets
- 2) the secondary wavelet spread out in all directions with the speed of light in the given medium

- 3) the new wavefront at any later time is given by the forward envelope, (tangential surface in the forward direction) of the secondary wavelets at that time





Reflection of light



Reflection on the basis of wave theory

Let angles of incidence and reflection be i and r respectively

$\angle BAC = \angle CDB$ (each is 90°)

$BC = BC$ (common)

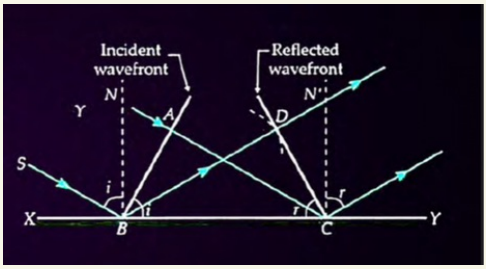
$AC = BD$ (each is equal to ct)

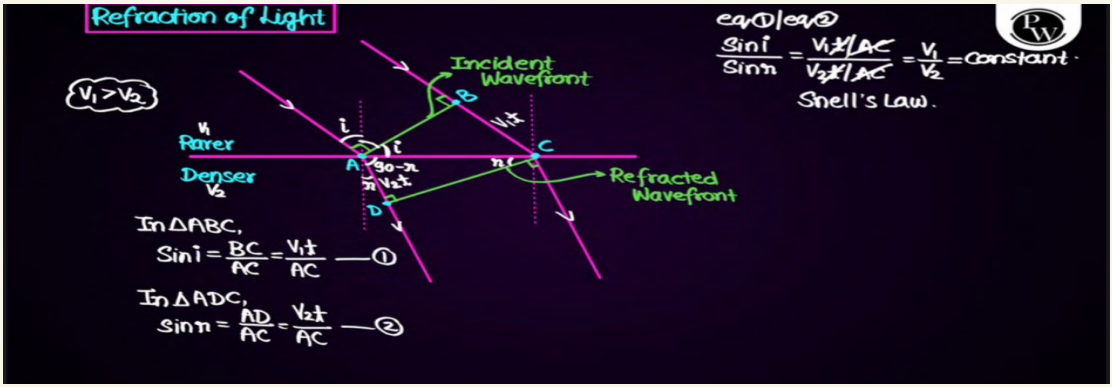
$\therefore \Delta ABC = \Delta DCB$

Hence, $\angle ABC = \angle DCB$

or $\angle i = \angle r$

i.e., The angle of incidence is equal to the angle of reflection
This proves the first law of reflection





Refraction on the basis of wave theory

Let the an angle of incidence and refraction be i and r respectively

From right ΔABC, we have

$$\sin \angle BAC = \sin i = \frac{BC}{AC}$$

From right ΔADC, we have

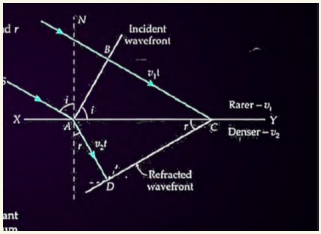
$$\sin \angle DCA = \sin r = \frac{AD}{AC}$$

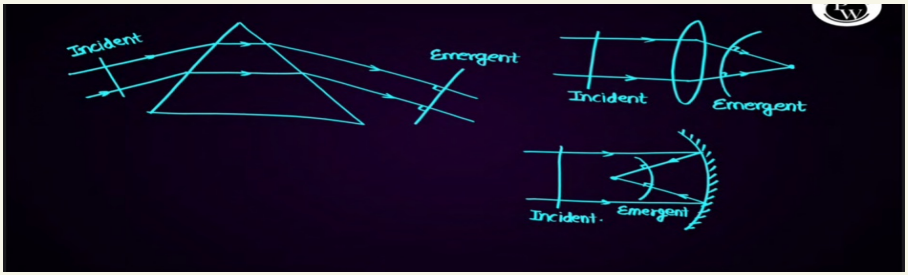
$$\therefore \quad \frac{\sin i}{\sin r} = \frac{BC}{AD} = \frac{v_1 t}{v_2 t}$$

or $$\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = {}^1\mu_2 \text{ (a constant)}$$

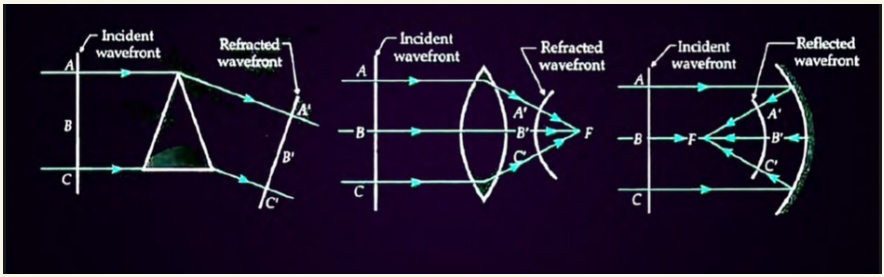
This proves Snells law of refraction

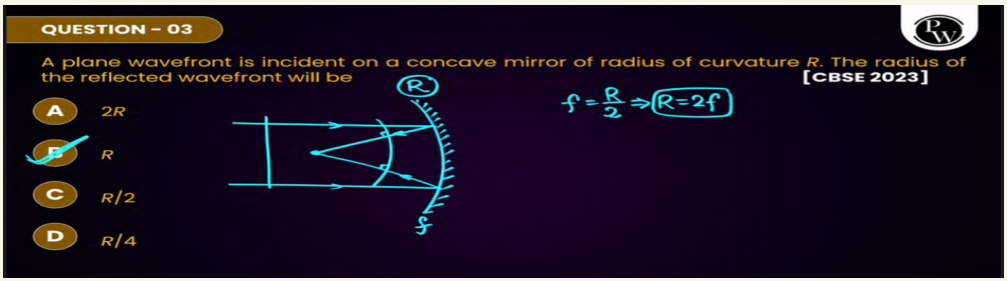
The constant is called the refractive index of the second medium with respect to the first medium





Behaviour of a prism, lens and mirror

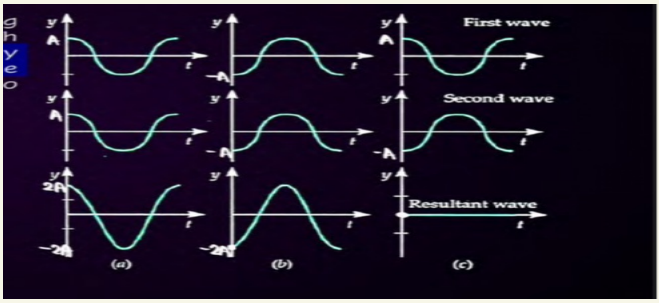




Principle of superimposition

$\vec{y} = \vec{y}_1 + \vec{y}_2 + \ldots$

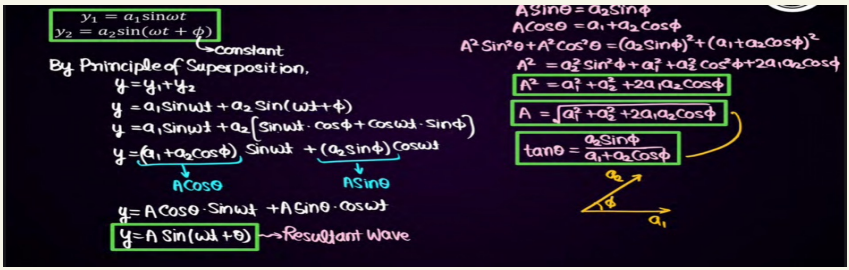
- when a number of waves travelling through a medium, super impose on each other, the resultant displacement at any point at a given instant is equal to the vector sum of the displacement due to the individual waves at that point

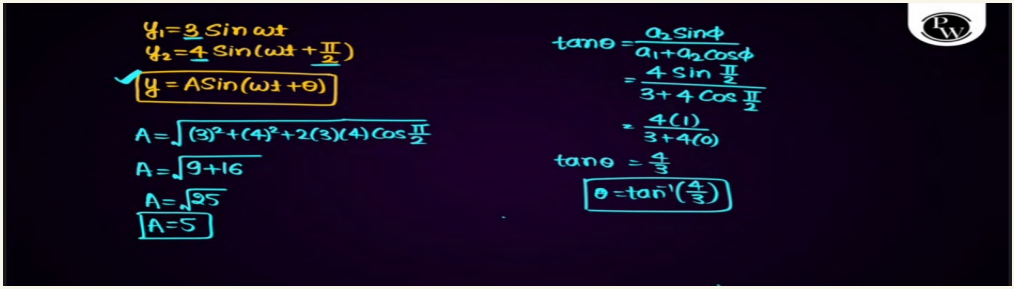


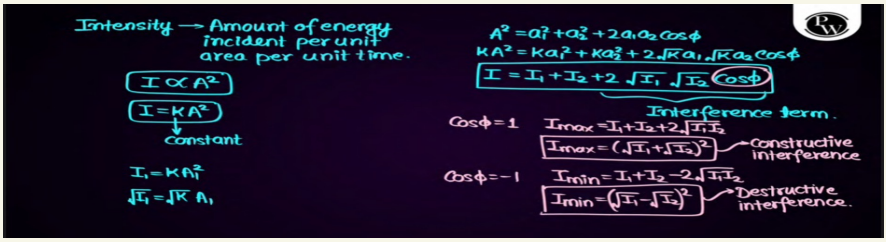
Interference of light: Young's double slit experiment

- when two light waves of the same frequency and having zero or constant phase difference, travelling in the same direction superpose on each other, the intensity in the region of super imposition gets re-distributed, becoming maximum at some point and minimum at others. This phenomena is called interference of light.

Conditions for constructive and destructive interference







QUESTION –04 (CBSE 2023)

In the wave picture of light, the intensity I of light is related to the amplitude A of the wave as: [1 Mark]

A $I \propto \sqrt{A}$ $I \propto A^2$

B $I \propto A$

C $I \propto A^2$

D $I \propto 1/A^2$

Relation between $\Delta\phi$ and Δp

$$\Delta\phi = \frac{2\pi}{\lambda}\Delta p$$

$$\Delta p = \frac{\lambda}{2\pi}\Delta\phi$$

Constructive interference

I_{max}

$\cos\phi = 1$

$\phi = 2n\pi$ $n = 0, 1, 2, \ldots$

even multiple of π

path difference

$\Delta p = \frac{\lambda}{2\pi}(2n\pi) = \Delta p = n\lambda$

$n = 0, 1, 2 \ldots$

Destructive interference

I_{min}

$\cos\phi = -1$

$\phi = (2n-1)\pi$ $n = 1, 2, 3, \ldots$

odd multiple of π

path diff

$\Delta p = \frac{\lambda}{2\pi}(2n-1)\pi$

$\Delta p = (2n-1)\frac{\lambda}{2}$ $n = 1, 2, 3, \ldots$

Two monochromatic light waves of amplitude A and $2A$ interfering at a point, have a phase difference of 60°. The intensity at that point will be proportional to :

A. $3A^2$
B. $5A^2$
C. $7A^2$
D. $9A^2$

$$A_R^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\phi$$
$$= A^2 + (2A)^2 + 2(A)(2A)\frac{1}{2}$$
$$= 5A^2 + 2A^2$$
$$= 7A^2$$

QUESTION – 06

The displacement of the interfering light waves are $y_1 = 4\sin\omega t$ and $y_2 = 3\sin(\omega t + \pi/2)$. What is the amplitude of the resultant wave?

A. 5
B. 7
C. 1
D. 0

$$A^2 = 4^2 + 3^2 + 2(4)(3)\cos\frac{\pi}{2}$$
$$= 16 + 9$$
$$A^2 = 25$$
$$A = 5$$

Coherent and incoherent sources

- two sources of light, which continuously emit light waves of same frequency with a zero or constant phase difference between them are called coherent sources.
- Two sources of light, which do not emit light with a constant phase difference are called incoherent sources

Ratio of intensity of light at maxima and minima

$$\frac{I_{max}}{I_{min}} = \frac{(\sqrt{I_1} + \sqrt{I_2})^2}{(\sqrt{I_1} - \sqrt{I_2})^2}$$

$$= \frac{\sqrt{(kA_1)^2 + (kA_2)^2}}{\sqrt{(kA_1)^2 - (kA_2)^2}}$$

$$\frac{I_{max}}{I_{min}} = \left[\frac{(A_1 + A_2)}{(A_1 - A_2)}\right]^2$$

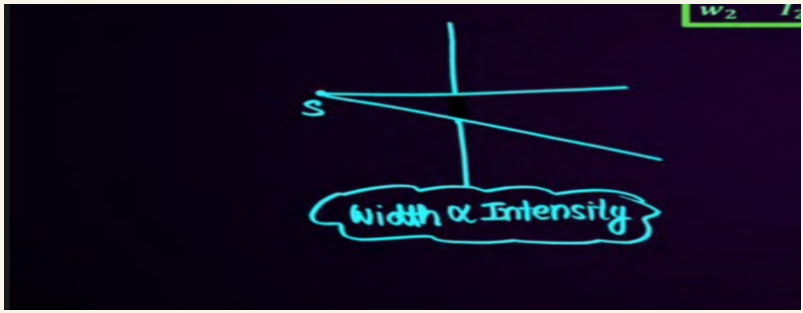
let $\frac{A_1}{A_2} = r$

$$\frac{I_{max}}{I_{min}} = \left(\frac{r+1}{r-1}\right)^2$$

Ratio of intensity of light due to 2 sources

The intensity of light through a slit is proportional to its width
If w1 and w2 are the width of the two slit s1 and s2 then

$$\frac{w_1}{w_2} = \frac{I_1}{I_2} = \frac{a_1^2}{a_2^2} = r^2$$



QUESTION – 07

Two coherent sources have intensities in the ratio 25:16. Find the ratio of the intensities of maxima to minima, after interference of light occurs.

$$\frac{I_1}{I_2} = \frac{25}{16}$$

$$\frac{KA_1^2}{KA_2^2} = \frac{25}{16}$$

$$n = \frac{A_1}{A_2} = \frac{5}{4}$$

$$\frac{I_{max}}{I_{min}} = \left(\frac{n+1}{n-1}\right)^2 = \left(\frac{\frac{5}{4}+1}{\frac{5}{4}-1}\right)^2 = \left(\frac{9/4}{1/4}\right)^2 = 81:1$$

QUESTION – 08

Two sources of intensity I and $4I$ are used in an interference experiment. Find the intensity at points where the waves from two sources superimpose with a phase difference (i) zero (ii) $\pi/2$ [CBSE]

(i) $I_R = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

$= I + 4I + 2\sqrt{I(4I)}\cos 0$

$= 5I + 2(2I)$

$I_R = 9I$

(ii) $I_R = 5I$

QUESTION – 09

When interference of light takes place :

A. energy is created in the region of maximum intensity

B. energy is destroyed in the region of maximum intensity

C. conservation of energy holds good and energy is redistributed ✓

D. conservation of energy does not hold good.

QUESTION – 10

The light sources used in Young's double slit experiment are:

A. incoherent

B. coherent ✓ $f \rightarrow$ same, $\phi \rightarrow 0$ or constant

C. white light

D. blue-green-red light

I_1 , I_2

$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

let $I_1 = I_2 = I_0$

$I = I_0 + I_0 + 2\sqrt{I_0 I_0}\cos\phi$

$I = 2I_0 + 2I_0\cos\phi$

$\boxed{I = 2I_0(1+\cos\phi)}$

$I = 2I_0\left(2\cos^2\frac{\phi}{2}\right)$

$\boxed{I = 4I_0\cos^2\frac{\phi}{2}}$

$\cos 2\theta = 2\cos^2\theta - 1$

$\cos\theta = 2\cos^2\frac{\theta}{2} - 1$

$1 + \cos\theta = 2\cos^2\frac{\theta}{2}$

QUESTION – 11

In a Young's double slit experiment, the intensity of light at a point on the screen where the path difference is λ is K units. Find the intensity at a point where the path difference is $\frac{\lambda}{4}$

let $I_1 = I_2 = I_0$

$\Delta\phi = \frac{2\pi}{\lambda}(\lambda) = 2\pi$

$I = 4I_0\cos^2\frac{\phi}{2}$

$K = 4I_0\cos^2\frac{2\pi}{2}$

$K = 4I_0(-1)^2$

$\boxed{K = 4I_0}$

$\Delta\phi = \frac{2\pi}{\lambda}\left(\frac{\lambda}{4}\right) = \frac{\pi}{2}$

$I' = 4I_0\cos^2\frac{\phi}{2}$

$= 4I_0\cos^2\frac{\pi}{4}$

$= 4I_0\left(\frac{1}{\sqrt{2}}\right)^2$

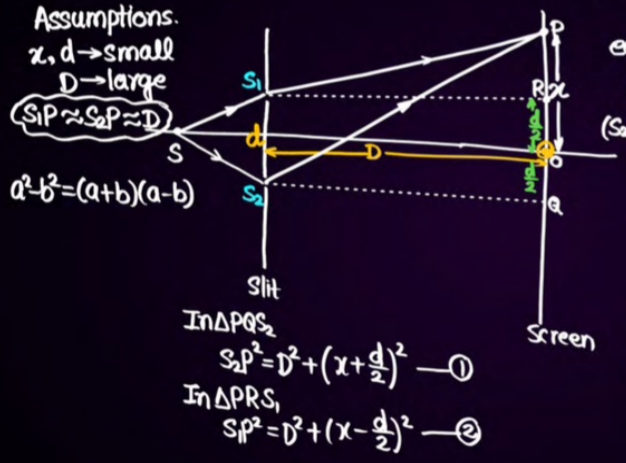
$= 2I_0$

$\boxed{I' = \frac{K}{2}}$

YDSE

Assumptions.
$x, d \rightarrow$ small
$D \rightarrow$ large
$S_1P \approx S_2P \approx D$

$a^2-b^2=(a+b)(a-b)$



In ΔPQS_2

$$S_2P^2 = D^2 + \left(x+\frac{d}{2}\right)^2 \quad -(1)$$

In ΔPRS_1

$$S_1P^2 = D^2 + \left(x-\frac{d}{2}\right)^2 \quad -(2)$$

Path Difference

$$\Delta p = S_2P - S_1P$$

eq(1) − eq(2)

$$S_2P^2 - S_1P^2 = \left(x+\frac{d}{2}\right)^2 - \left(x-\frac{d}{2}\right)^2$$

$$(S_2P - S_1P)(S_2P + S_1P) = (2x)d$$

$$\Delta p = \frac{2xd}{S_2P+S_1P}$$

$$\Delta p = \frac{2xd}{D+D}$$

$$= \frac{2xd}{2D}$$

$$\boxed{\Delta p = \frac{xd}{D}}$$

For constructive interference,

$$\Delta p = n\lambda \qquad n = 0,1,2\ldots$$

$$\frac{xd}{D} = n\lambda$$

$$\boxed{x = \frac{nD\lambda}{d}}$$

Positions of Maxima

$$x = 0, \frac{D\lambda}{d}, \frac{2D\lambda}{d}, \frac{3D\lambda}{d} \ldots$$

For destructive interference,

$$\Delta p = (2n-1)\frac{\lambda}{2} \qquad n = 1,2,3\ldots$$

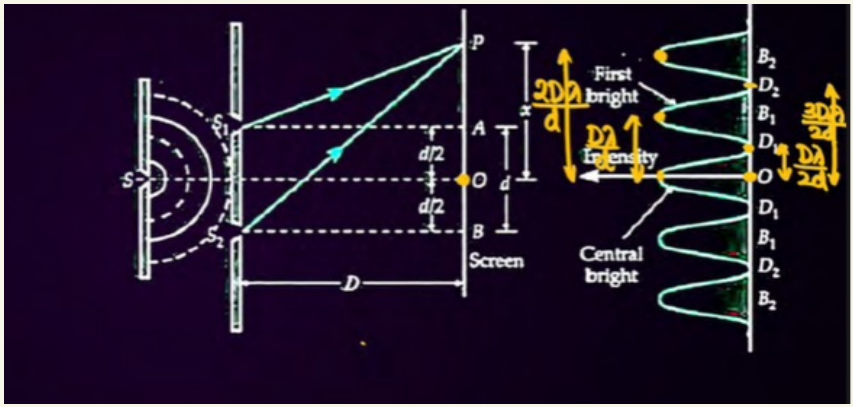
$$\frac{xd}{D} = (2n-1)\frac{\lambda}{2}$$

Positions of minima

$$\boxed{x = \frac{(2n-1)D\lambda}{2d}}$$

$$x = \frac{D\lambda}{2d}, \frac{3D\lambda}{2d}, \frac{5D\lambda}{2d}, \ldots$$

Theory of interference fringes: fringe width



Position of bright fringes

$$p = \frac{xd}{D} = n\lambda$$

$$x = \frac{nD\lambda}{d} \qquad n = 0, 1, 2, 3 \ldots$$

Clearly, the positions of various bright fringes are as follows

For, $n = 0$, $x_0 = 0$ Central bright fringes

For, $n = 1$, $x_1 = \frac{D\lambda}{d}$ First bright Fringe

For, $n = 2$, $x_2 = \frac{2D\lambda}{d}$ Second bright fringe

For, $n = n_t$ $x_n = \frac{nD\lambda}{d}$ n_{th} Bright fringe

Position of dark fringes

For destructive interference

$$p = \frac{xd}{D} = (2n-1)\frac{\lambda}{2}$$

or $\boxed{x = (2n-1)\frac{D\lambda}{2d}}$ $n = 1, 2, 3, \ldots$

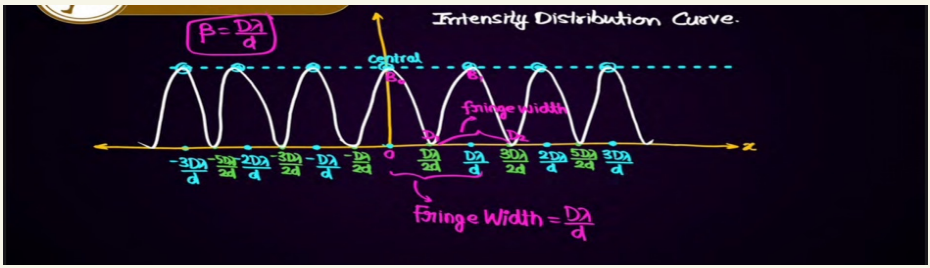
Clearly, the positions of various dark fringes are as follows

For, $n = 1$, $x_1' = \frac{1}{2}\frac{D\lambda}{d}$ First dark fringe

For, $n = 2$, $x_2' = \frac{3}{2}\frac{D\lambda}{d}$ Second dark fringe

For, $n = n_t$ $x_n' = \frac{(2n-1)D\lambda}{d}$ nth Dark fringe

Fringe width



- if Young's double slit apparatus is emerged in a liquid of refractive index μ the wavelength of light decreases to. $\lambda' (= \lambda/\mu)$ And so the fringe width reduces.

$$\boxed{\beta' = \frac{D\lambda'}{d} = \frac{D\lambda}{\mu d} = \frac{\beta}{\mu}}$$

QUESTION –12 (CBSE 2024)

In a Young's double-slit experiment in air, the fringe width is found to be 0.44 mm. If the entire setup is immersed in water $\left(n = \frac{4}{3}\right)$, the fringe width will be [1 Mark]

β

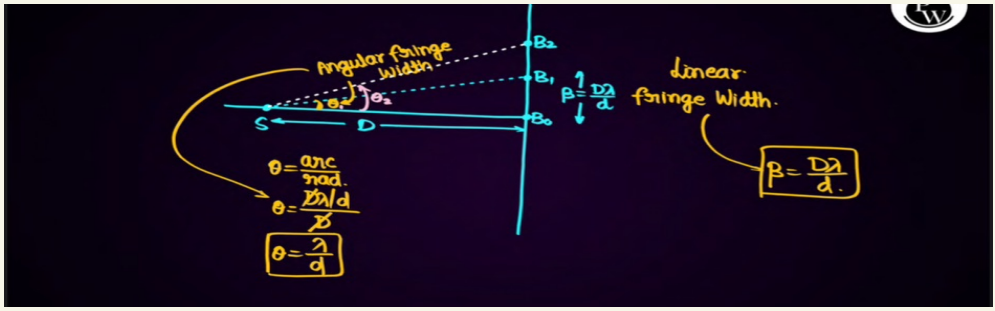
$\beta' = \frac{\beta}{\mu} = \frac{0.44\,(3)}{4} = 0.33\text{mm}$.

A 0.88 mm

B 0.59 mm

C 0.33 mm

D 0.44 mm



QUESTION –13 (CBSE 2023)

In a Young's double-slit experiment, the screen is moved away from the plane of the slits. What will be its effect on the following?
(i) Angular separation of the fringes.
(ii) Fringe-width. [1 Mark]

$\theta = \frac{\lambda}{d}$ $\beta = \frac{D\lambda}{d}$ $D\uparrow$

4000 Å - 7000 Å

A Both (i) and (ii) remain constant.

B (i) remains constant, but (ii) decreases.

C (i) remains constant, but (ii) increases.

D Both (i) and (ii) increase.

QUESTION – 14

In Young's experiment, the width of the fringes obtained with light of wavelength 6000Å is 2.0 mm. Calculate the fringe width if the entire apparatus is immersed in a liquid medium of refractive index 1.33. [CBSE]

$\beta = 2mm$

$\eta = 1.33$

$\beta' = \frac{2}{1.33}$ mm

QUESTION – 15 (Imp)

A beam of light consisting of two wavelengths, 800 nm (λ_1) and 600 nm (λ_2) is used to obtain the interference fringes in a Young's double slit experiment on a screen placed 1.4 m away. If the two slits are separated by 0.28 mm, calculate the least distance from the central bright maximum where the bright fringes of the two wavelengths coincide.

$D = 1.4m$.

$d = 0.28mm$.

n^{th} Bright fringe of λ_1 Coincides with $(n+1)^{th}$ bright fringe of λ_2

Distance of 3rd bright fringe of λ_1

$x_3 = \frac{3D\lambda}{d}$

$= \frac{3 \times 1.4 \times 800}{0.28 \times 10^{-3}}$

$= 12 mm$.

$x_n = x_{n+1}$

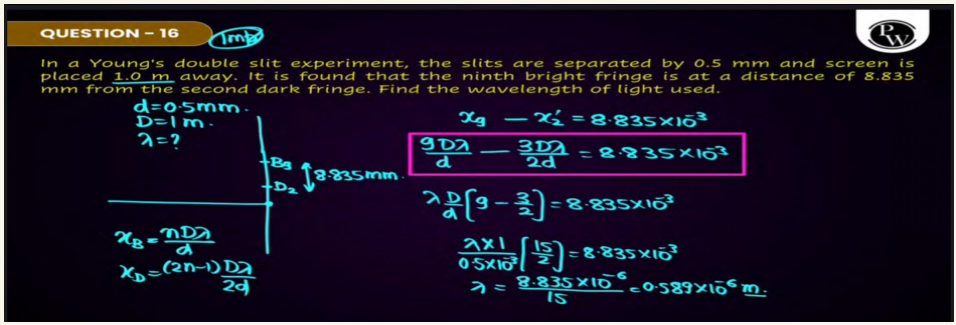
$\frac{n\cancel{B}\lambda_1}{\cancel{d}} = \frac{(n+1)\cancel{B}\lambda_2}{\cancel{d}}$

$n(8000) = (n+1)6000$

$8n = 6n + 6$

$2n = 6$

$n = 3$



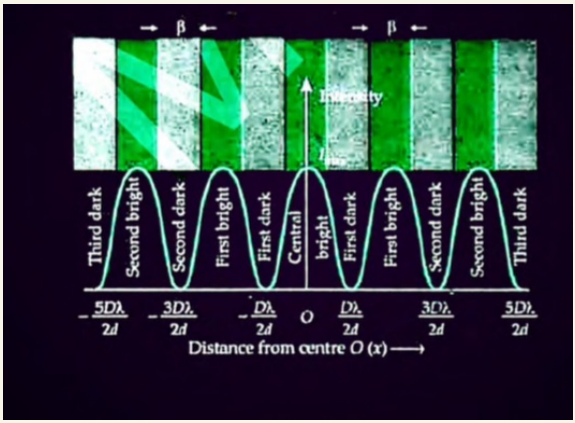
Sustained interference

- the interference pattern in which the positions of maximum and minimum of intensity of light remain fixed all along on the screen is called sustained or permanent interference pattern

Conditions for sustained interference

- the two sources should emit the light waves continuously
- The light wave should be of same wavelength
- The light waves emitted should preferably be of the same amplitude. If the amplitude of the waves are equal the dark fringes are completely dark.
- The waves emitted by the two sources of light should either be in phase or should have a constant phase difference.

Intensity distribution curve for interference



Interference pattern with white light

- white light consist of colours from violet to red with wavelength range from 4000 A to 7000 A. Different component Colours of white light produce their own interference pattern.
- at the centre of the screen, the path difference is zero for all such components, so bright fringes of different colour overlap at the centre. Consequently, the central fringe is white.
- Now fringe width $\beta = D\lambda/d$ i.e, $\beta \propto \lambda$ Since the violet colour has the lowest wavelength, the closest fringe to either side on the Central white fringe is violet, while the farthest fringe is red.
- After a few fringes, the interference pattern is lost due to large overlapping of the fringes and uniform white illumination is seen on the screen

QUESTION –17 (CBSE 2024) $\beta = \frac{D\lambda}{d}$ $\beta \to \infty$

Assertion (A): In a Young's double-slit experiment, interference pattern is not observed when two coherent sources are infinitely close to each other.
Reason (R): The fringe width is proportional to the separation between the two sources. $\beta \propto \frac{1}{d}$
In the light of the above statements, choose the correct answer from the options given below: [1 Mark]

A. Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).

B. Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).

C. Assertion (A) is true and Reason (R) is false.

D. Assertion (A) is false and Reason (R) is also false.

QUESTION –18 (CBSE 2024) Imb

A Young's double-slit experimental set up is kept in a medium of refractive index $(\frac{4}{3})$. Which maximum in this case will coincide with the 6th maximum obtained if the medium is replaced by air? [1 Mark]

A. 4th

B. 6th

C. 8th

D. 10th

$$\frac{6\beta\lambda_{air}}{d} = \frac{n\beta\lambda_{med.}}{d} \qquad n = \frac{\lambda_a}{\lambda_m}$$

$$6\lambda_a = n\lambda_m.$$

$$n = 6\frac{\lambda_a}{\lambda_m}$$

$$n = 6 \times \frac{4}{3} = 8$$

QUESTION – 20

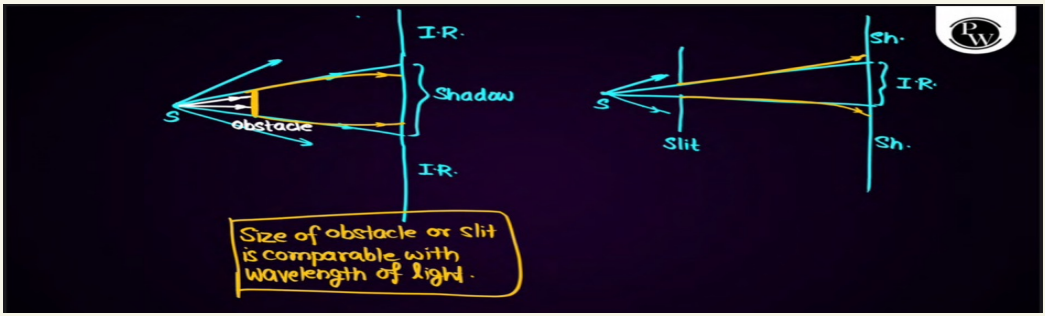
In a Young's double-slit experiment the source is white light. One of the holes is covered by a red filter and another by a blue filter. In this case,

(A) there shall be alternate interference patterns of red and blue.

(B) there shall be an interference pattern for red distinct from that for blue.

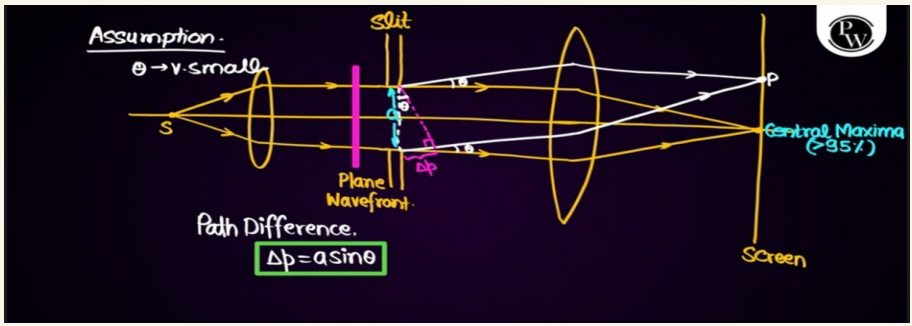
(C) there shall be no interference fringes.

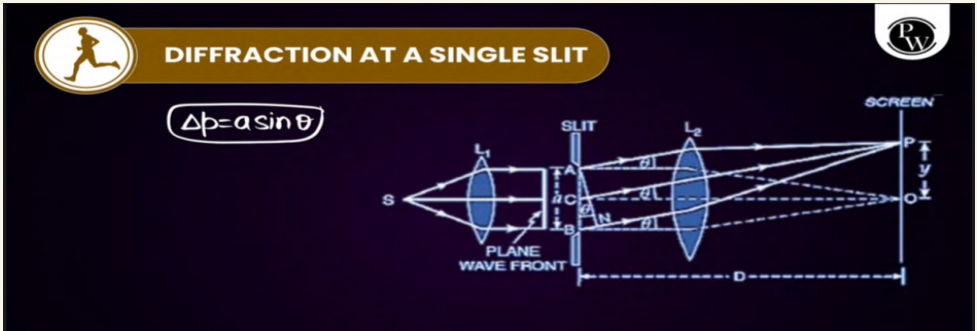
(D) there shall be an interference pattern for red mixing with one for blue.

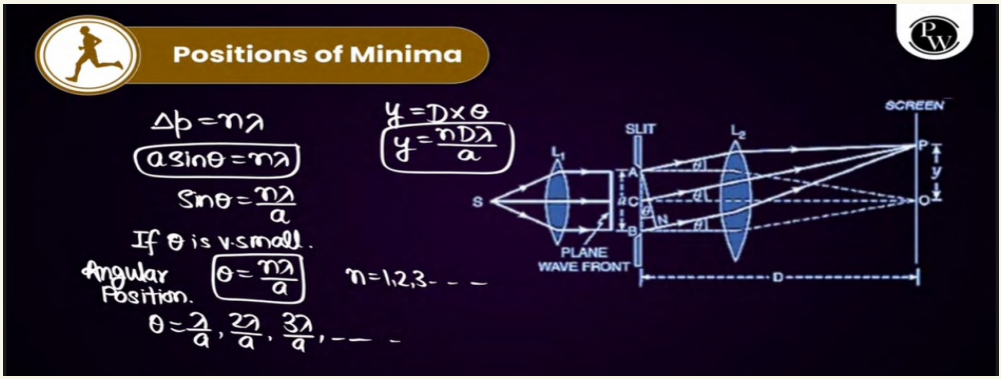


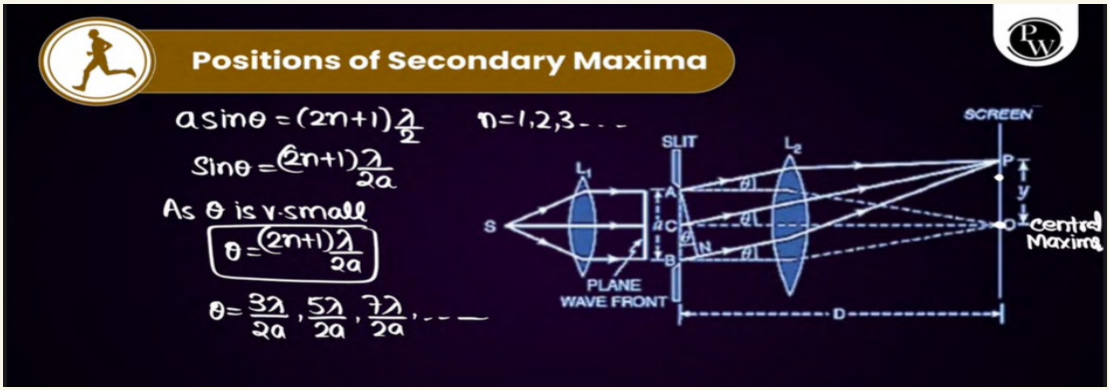
Diffraction of light

- the phenomena of bending of light around the corners of small obstacles or apertures, and it consequence spreading into the regions of geometrical shadow is called diffraction of light







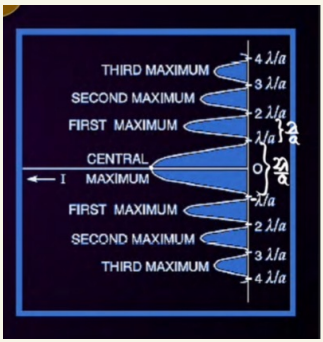


Intensity distribution curve

- the intensity of secondary maxima relating to the intensity of the central maximum are in ratio

$$1 : \frac{1}{21} : \frac{1}{61} : \frac{1}{121} : \ldots$$

- The intensity of the first secondary maximum is just 4% of that of the central maximum



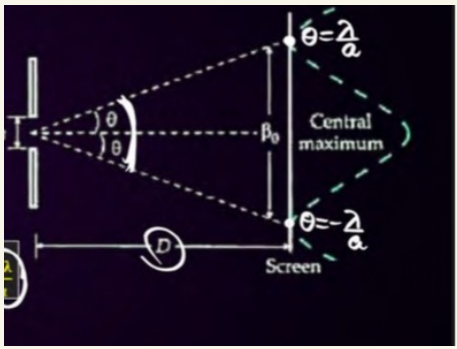
Width of central and secondary maxima

Angular width of central Maxima

- the direction of the first minima on either side of central Maxima can be given by

$$\theta = \frac{\lambda}{a}$$

- this angle is called half angular width of central maxima
- Angular width of central maximum $= 2\theta = \frac{2\lambda}{a}$



QUESTION –21 (CBSE 2023)

In a single-slit diffraction experiment, the width of the slit is halved. The width of the central maximum, in the diffraction pattern, will become: [1 Mark]

A) half
B) twice
C) four times
D) one-fourth

$\theta = \frac{2\lambda}{a}$

$\theta' = \frac{2\lambda}{(\frac{a}{2})} = 2\theta$

Difference between Interference and Diffraction

Basis of Difference	Interference	Diffraction
Fringe width	Fringe width is constant.	Fringe width varies.
Source fringes	Fringes are obtained with the coherent light coming from two slits.	Fringes are obtained with the monochromatic light coming from single slit.
Dependency	It depends upon the distance between two openings.	It depends upon the aperture of single slit opening.

Difference between Interference and Diffraction

Basis of Difference	Interference	Diffraction
Visibility	Many fringes are visible.	Fewer fringes are visible.
Brightness	All fringes are of same brightness.	Central fringe has maximum reduces gradually.
Slit or obstacle	Not required	required

QUESTION – 22

(Imp)

A parallel beam of light of wavelength 600 nm is incident normally on a slit of width 'a'. If the distance between the slits and the screen is 0.8 m and the distance of 2nd order maximum from the centre of the screen is 15 mm calculate the width of the slit.

$\lambda = 600\,nm.$

$a = ?$

$D = 0.8\,m.$

$$x = \frac{(2n+1)D\lambda}{2a}$$

$n = 2 \Rightarrow x = \frac{5D\lambda}{2a} = 15 \times 10^{-3}$

$$a = \frac{5D\lambda}{30 \times 10^{-3}} = \frac{5 \times 0.8 \times 600 \times 10^{-9}}{30 \times 10^{-3}} = 80 \times 10^{-6} = 8 \times 10^{-5}\,m.$$

QUESTION – 23

Fraunhoffer diffraction from a single slit of width 1.0μm is observed with light of wavelength 500nm Calculate the half angular width of the central maximum. [ISC]

$a = 1\,\mu m.$

$\lambda = 500\,nm.$

$$\theta = \frac{2\lambda}{a}$$

$$\frac{\theta}{2} = \frac{\lambda}{a} = \frac{500 \times 10^{-9}}{10^{-6}}$$

$$= 500 \times 10^{-3}$$

QUESTION – 24

Light of wavelength 600nm falls normally on a slit of width 1.2μm producing Fraunhoffer diffraction pattern on a screen. Calculate the angular position of the first minimum and the angular width of the central maximum.

$\lambda = 600\,nm.$

$a = 1.2\,\mu m.$

$$\theta = \frac{\lambda}{a} = \frac{600 \times 10^{-9}}{1.2 \times 10^{-6}}$$

Angular Width $= \frac{2\lambda}{a}.$

QUESTION – 25

A slit of width 'a' is illuminated by red light of wavelength 6500Å For what value of 'a' will (i) the first minimum fall at an angle of diffraction of 30° and (ii) the first maximum fall at an angle of diffraction of 30°. [CBSE]

Solution. (i) For first minimum of the diffraction pattern,

$$a\sin\theta = \lambda$$

$$a = \frac{\lambda}{\sin\theta} = \frac{6{,}500 \times 10^{-10} m}{\sin 30°}$$

$$\therefore\ a = \frac{6{,}500 \times 10^{-10} m}{0.5} = 1.3 \times 10^{-6} m.$$

(ii) For first secondary maximum of the diffraction pattern,

$$a\sin\theta = \frac{3\lambda}{2}$$

$$\therefore\ a = \frac{3\lambda}{2\sin\theta} = \frac{3 \times 6{,}500 \times 10^{-10} m}{2 \times \sin 30°}$$

$$= 1.95 \times 10^{-6} m.$$

QUESTION – 26

(HW)

For what distance is ray optics a good approximation when the aperture is 3mm wide and the wavelength is 500nm ? [NCERT]

Solution. Here $a = 3mm = 3 \times 10^{-3} m,$

$$\lambda = 500nm = 500 \times 10^{-9} m$$

The distance upto which a beam of light can travel without significant broadening is called Fresnel distance and its value is given by

$$\boxed{D_F = \frac{a^2}{\lambda}} = \frac{(3 \times 10^{-3})^2}{500 \times 10^{-9}} = 18m.$$

QUESTION – 27

When a wave undergoes reflection at an interface from rarer to denser medium, then change in its phase is [All India 2020]

A $\pi/2$

B Zero

C π

D $\pi/4$

QUESTION – 28

Angular width of central maxima of a single slit diffraction pattern is independent of

$\theta = \frac{2\lambda}{a}$ $\quad c = f\lambda$

(A) slit width

(B) frequency of the light used

(C) wavelength of the light used

(D) distance between slit and screen (D)

QUESTION – 29

Why does the light appear to travel in a straight line in spite of its wave nature?

Ans. It is due to high frequency and small wavelength of light.

QUESTION – 30

Can we observe interference maxima or minima on a screen if the slits are separated by a distance less than a wavelength of light is used ?

Ans. No, single fringe will cover the whole screen.

$d \simeq \lambda$

QUESTION – 31

Why we cannot get diffraction from a wide slit illuminated by a monochromatic light?

Ans. The size of the slit should be of order of wavelength of light to observe it's diffraction.

QUESTION – 32

State with reason whether diffraction of light takes place at the two slits in the Young's interference experiment.

Ans. Yes, in YDSE, the fringe pattern is the result of superposition between two individual diffraction patterns from the two slits, so diffraction is in background.

(a) Wave front :
The locus of all the particles of a medium vibrating in the same phase at any given instant.
The shape of a wave front depends upon the shape of the source light.

(b) Types of wave front :

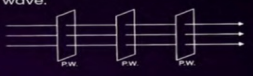
(i) Spherical wave front: Spherical wave front is produced, if the source of disturbance is a point source.



(ii) Cylindrical wave front : Cylindrical wave front is produced, if the source of disturbance is a slit (line source).



(iii) Plane wave front : A plane wave front is normal to the direction of propagation of wave.



Note :
The direction of light rays are always perpendicular to the wave front and in outwards direction from the source.

HUYGEN'S PRINCIPLE :

According to Huygen's Principle :

(i) Every point on a wave front acts like a light source and emits secondary wavelets. These secondary wavelets spread in all direction in a given medium with a velocity of light.

(ii) The envelope of wave front of secondary wavelets, moves along forward direction gives new position of wave front.

Note:

(a) Huygen's principle explains the phenomenon of diffraction, polarisation of interference of light.

(b) The laws of reflection and refraction can be verified using Huygen's principle of geometrical construction.

LAW OF REFLECTION ON THE BASIS OF HUYGEN'S WAVE THEORY :



Let AB is an incident plane wave front and CD is a reflected wave front. Here, $BD = AC = ct$, where c is the speed of wave front and t is the time taken by the disturbance to travel from A to C.
Let angle of incident and reflected are i and r respectively. In $\triangle ABC$ and $\triangle DCB$, we have

$\angle BAC = \angle CDB$ [each 90°]
$BC = BC$ [common]
$AC = BD$ [equal to ct]

$\therefore \Delta ABC \cong \Delta DCB$
$\Rightarrow \angle ABB = \angle DCB$
$i = r$

This prove the law of reflection of light.

LAW OF REFRACTION ON THE BASIS OF HUYGEN'S WAVE THEORY :



Let c be the speed of incident wave front in rare medium and v be the speed of refracted wave front in denser medium.
Here, $BC = ct$ $AD = vt$
AB is an incident wave front in rare medium and CD is refractive wave front in denser medium.

From right $\triangle ABC, \sin i = \frac{BC}{AC}$

From right $\triangle ADC, \sin r = \frac{AD}{AC}$

$\therefore \frac{\sin i}{\sin r} = \frac{BC/AC}{AD/AC} = \frac{c}{v} = \mu$ (a constant)

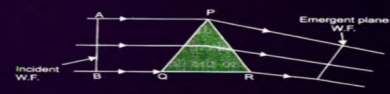
This proves Snell's law of refraction.

REFLECTION AND REFRACTION OF PLANE WAVE FRONT :

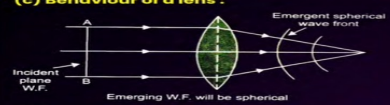
(a) Behaviour of a concave mirror :



(b) Behaviour of a prism :



(c) Behaviour of a lens :



SUPERPOSITION PRINCIPLE :

This principle states that the resultant displacement of a particle at any instant is the vector sum of the individual displacements produced by each wave.

$$\mathbf{y} = \mathbf{y_1} + \mathbf{y_2} + \mathbf{y_3} + \cdots$$

Note:

(a) Constructive superposition : When two waves of same wavelengths superimpose on each other in phase i.e., crest of one wave falls on the crest of other wave or trough of one wave falls on the trough of other wave.

$$\bar{y} = \overline{y_1} + \overline{y_2}$$

(b) Destructive superposition : When two waves of same wavelengths superimpose on each other. Out of the phase the crest of one wave falls on the trough of the other wave and vice-versa, then the superposition is destructive.

$$\bar{y} = \overline{y_1} - \overline{y_2}$$

(a) Coherent sources of light : Two sources of light are said to be coherent, if they emit waves of same frequency and are either in same phase or have a constant phase difference.

(b) Incoherent sources of light : Two sources having different frequencies or the source have same frequency but no constant phase difference are called incoherent sources.

INTERFERENCE OF LIGHT :

The phenomenon of redistribution of light energy in the region of superposition of wave is called interference of light.

Note:

(a) The point where the intensity of light is maximum due to superposition of waves are said to be constructive interference.

(b) The point where the intensity of light is minimum due to superposition of waves are said to be destructive interference.

(c) Two waves of amplitudes a_1 and a_2 and let ϕ be the phase difference, then the resultant amplitude of the superposition of waves is given by

$$A^2 = a_1^2 + a_2^2 + 2a_1a_2\cos\phi$$

For constructive interference, $A = A_{max}$

$$\therefore A_{max} = (a_1 + a_2)^2$$

For destructive interference, $A = A_{min}$

$$A_{min} = (a_1 - a_2)^2$$

Intensity of light is directly proportional to the square of the amplitude

$$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$$

If $I_1 = I_2 = I_0$, then resultant intensity of light is given by

$$I_R = I_0 + I_0 + 2I_0\cos\phi = 2I_0(1 + \cos\phi)$$

$$\therefore I_R = 4I_0\cos^2(\phi/2)$$

(d) Conditions for constructive interference :

(i) Phase difference $(\phi) = 2\pi n$, **where** $n = 1, 2, 3, \ldots$

(ii) Path difference $(\Delta x) = n\lambda$, **where** $n = 1, 2, 3, \ldots$

$$I_{max} \propto (a_1 + a_2)^2$$

(e) Conditions for destructive interference :

(i) Phase differecne $(\phi) = (2\pi - 1)\pi, n = 1, 2, 3, \ldots$

(ii) Path difference $(\Delta x) = (2n - 1)\frac{\lambda}{2}$, $n = 1, 2, 3 \ldots$

(f) (i) Path difference $= \frac{\lambda}{2\pi} \times$ **Phase difference**

(ii) Phase difference $(\phi) = \frac{2\pi}{\lambda} \times$ **Path difference**

YOUNG'S DOUBLE SLIT EXPERIMENT :

Dark and bright bands in the interference pattern are called interference fringes.

(a) Fringe width of bright and dark fringe is given by

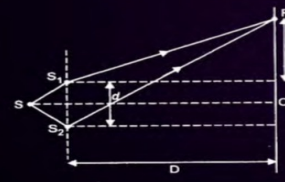
$$\beta = \frac{\lambda D}{d}$$

Where

β = fringe width

λ = wavelength of wave

D = distance between source and screen



(b) Separation of nth bright fringe from central fringe

$$y_n = \frac{n\lambda D}{d}, \text{ where } n = 1, 2, 3, \ldots$$

(c) Separation of nth dark fringe from central fringe

$y_n = (2n - 1)\frac{\lambda D}{2d}$, where $n = 1, 2, 3, \ldots$

(d) Angular fringe width (for dark and bright) is given by $\theta = \frac{\beta}{D} = \frac{\lambda}{d}$

(e) Angular position of nth order

(i) Bright fringe $= \frac{y_n}{D} = \frac{n\lambda}{d}$

(ii) Dark fringe $= \frac{y_n}{D} = (2n - 1)\frac{\lambda}{d}$

Note :

Factors on which fringe width depends

$$\beta = \frac{\lambda D}{d}$$

(i) $\beta \propto \lambda$, fringe width will be greater for red colour than for violet colour.

(ii) $\beta \propto D$

(iii) $\beta \propto \frac{1}{d}$

(iv) Fringe width decreases, when whole apparatus is taken from air to a denser medium, due to the decrease in wavelength of the light.

(v) $\theta = \frac{\beta}{D} = \frac{\lambda D}{dD}$

$\theta = \frac{\lambda}{d}$ [θ = angular fringe width]

(a) Angular width decreases with the increase in the distance between the coherent sources or vice-versa.

(b) Angular width of a fringe increase with increase in the wavelength λ of light used.

SUSTAINED INTERFERENCE OF LIGHT :

If the position of bright fringes and dark fringes remains fixed on the screen such a interference is called sustained interference. Conditions for sustained interference of light:

(i) The two sources must be coherent.

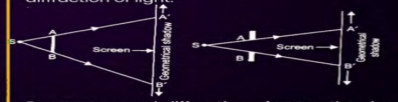
(ii) The two coherent sources of light should be narrow.

(iii) The distance between coherent sources of light and screen must be large.

(iv) The amplitudes of two waves emitted by the two coherent sources should be same.

DIFFRACTION OF LIGHT :

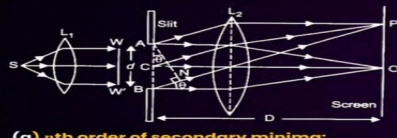
The phenomenon of bending of light at the corners of an object into the region of geometrical shadow of object is called diffraction of light.



For pronounced diffraction of wave, the size of the object should be of the order of the wavelength of the wave.

DIFFRACTION OF LIGHT DUE TO A SINGLE SLIT :

A plane wave front WW' is incident on the slit AB of width 'd'. The width of slit is of the order of wavelength of light used, so diffraction takes place through the slit.



(a) nth order of secondary minima:

When $d\sin\theta_n = n\lambda, n = 1,2,3, \ldots$ for small angle

$$\sin\theta \simeq \theta_n$$

$$\therefore d\theta_n = n\lambda$$

$\theta_n = \frac{n\lambda}{d}$ [θ_n = angular separation]

(b) nth order of secondary maxima:

$$d\sin\theta_n = (2n + 1)\frac{\lambda}{2}$$

$$\therefore \theta_n = (2n + 1)\frac{\lambda}{2d}$$

(c) Linear separation of nth secondary minima $y_n = \frac{n\lambda D}{d}$

(d) Angular width of central maxima $= \frac{2\lambda}{d}$

(e) Linear width of central maxima $= \frac{2\lambda D}{d}$

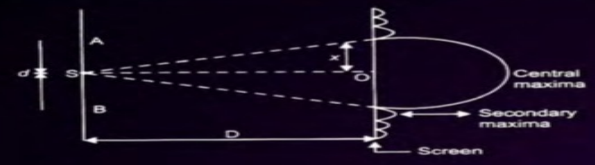
(f) Angular width of secondary maxima or minima $= \frac{\lambda}{d}$

(g) Linear width of secondary maxima or minima

$$(\beta) = \frac{\lambda D}{d} \Rightarrow \beta_0 = 2\beta$$

Linear width of central maxima is equal to two times of the linear width of secondary maxima or minima.

(h) The intensity of central maxima is maximum while the intensity of secondary maxima decrease with the increase of their order.



(i) Factors on which width of central maxima depends :

(a) The width of central maximum $\propto \lambda$, therefore, width of central maximum large for red colour and small for violet colour.

(b) The width of central maximum $\propto D$, so the width will increase with increase in D.

(c) The width of central maximum is inversely proportional to width (d) of the slit.

Wave Optics

Interference of Light

$y = y_1 + y_2$

$A^2 = A_1^2 + A_2^2 + 2A_1A_2\cos\phi$

$I = KA^2$

$I = I_1 + I_2 + 2\sqrt{I_1 I_2}\cos\phi$

If $I_1 = I_2 = I_0$

$I = 4I_0\cos^2\frac{\phi}{2}$

Phase Difference

$\Delta\phi = \frac{2\pi}{\lambda}\Delta p$

Constructive Interference

$\Delta\phi = 2n\pi$, $n = 0,1,2\cdots$

$\Delta p = n\lambda$

$I_{max} = (\sqrt{I_1} + \sqrt{I_2})^2$

Destructive Interference

$\Delta\phi = (2n-1)\pi$, $n = 1,2,3\cdots$

$\Delta p = (2n-1)\frac{\lambda}{2}$

$I_{min} = (\sqrt{I_1} - \sqrt{I_2})^2$

Intensity Ratio

$\frac{I_{max}}{I_{min}} = \left(\frac{A_1+A_2}{A_1-A_2}\right)^2 = \left(\frac{n+1}{n-1}\right)^2$

Relation with Slit Width

$\frac{W_1}{W_2} = \frac{I_1}{I_2} = \frac{A_1^2}{A_2^2}$

Position of Bright Fringe

$y = \frac{nD\lambda}{d}$ $n = 1,2,3\cdots$

Position of Dark Fringe

$y = (2n-1)\frac{D\lambda}{2d}$

Fringe Width

$\beta = \frac{D\lambda}{d}$, $\theta = \frac{\lambda}{d}$

Diffraction of Light

$\Delta p = d\sin\theta$

Position of secondary Minima

$d\sin\theta = n\lambda$ $n = 1,2\cdots$

Position of Secondary Maxima

$d\sin\theta = (2n+1)\frac{\lambda}{2}$, $n = 1,2\cdots$

Width of Central Maxima

$\beta = \frac{2D\lambda}{d}$

$\theta_0 = \frac{2\lambda}{d}$

Width of Sec. Maxima

$\beta = \frac{D\lambda}{d}$

$\theta = \frac{\lambda}{d}$

Fresnel's Distance

$D_F = \frac{a^2}{\lambda}$